Unbinned OmniFold measurement of the MINERvA ME FHC CC-inclusive double-differential cross section $d^2\sigma/(dp_T\, dp_\parallel)$

MINERvA internal technote — DRAFT

J. R. Bailey [TODO: collaborators]

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Abstract

We apply *unbinned* OmniFold to the MINERvA medium-energy FHC charged-current inclusive cross section, replacing the iterative Bayesian unfolding of the published analysis (arXiv:2106.16210). In two dimensions, $d^2\sigma/(dp_T\, dp_\parallel)$, our result reproduces the published spectrum (σ_{tot} ratio 1.011; 94 % of bins within 10 %) with a matching uncertainty budget (median relative σ 6.87 % vs 6.86 %).

Because adding an observable is just adding a feature—with no migration matrix or binning penalty—we then perform the first *simultaneous three- and four-observable* unfolds of MINERvA data, adding available energy (E_{avail}) and three-momentum transfer (q_3). Each higher-dimensional result reproduces the lower-dimensional one under marginalisation (a built-in closure), and four generators we produce at truth level (GENIE, the MINERvA Tune, NuWro, GiBUU) all under-predict the data at low E_{avail}. This is the well-established MINERvA low-recoil/2p2h deficit (Rodrigues 2016; Ascencio 2022): pion final-state interactions move $d\sigma/dE_{\mathrm{avail}}$ by $< 1\,\%$, whereas adding the empirical Valencia 2p2h fills the quasielastic–Δ dip. Recovering this known feature through an independent method validates the technique; the four-dimensional $(E_{\mathrm{avail}}, q_3)$ result additionally resolves the deficit's momentum-transfer structure and is set up for a bin-identical comparison to the published Ascencio measurement (arXiv:2110.13372).

The 2D and 3D results carry full covariance budgets (universe + bootstrap + ML), and the method passes binned and unbinned goodness-of-fit tests plus scalar GBDT/MLP classifier cross-checks. A point-cloud PET pipeline is implemented but not yet a validated physics result because the reconstructed-cluster source branch must be corrected and rerun. This note documents the method, the analysis pipeline, the uncertainty quantification, the validation suite, and the higher-dimensional extension.

Contents

1 Introduction and motivation

The MINERvA medium-energy (ME) FHC charged-current (CC) inclusive double-differential cross section in muon transverse and longitudinal momentum, $d^2\sigma/(dp_T\,dp_{||})$, was published by Ruterbories *et al.* [1] using D'Agostini iterative Bayesian unfolding (IBU) [2] on a binned response matrix. This note reproduces that measurement with *unbinned* OmniFold [3, 4], an unfolding technique based on iterated reweighting with machine-learned likelihood ratios that operates on per-event observables rather than on a fixed histogram binning. First demonstrated experimentally by the H1 collaboration [5, 6], OmniFold is one of a family of machine-learning unfolding methods [7–11] that lift the binned-response limitations of classical IBU, the SVD method [12] and TUnfold [13] (the latter accessed here via RooUnfold [14]).

Why unbinned OmniFold.

- **No binning at unfold time.** OmniFold learns event weights in the full observable space; the result is histogrammed only at the end, so the binning can be chosen (or refined) after unfolding without re-running.
- **Natural path to higher dimensions.** Adding observables is adding input features, not multiplying response-matrix bins — this is the enabling property for the available-energy extension (§7) that has no practical IBU analogue.
- **Different regularisation.** OmniFold's regulariser is the iteration count and the model capacity of the gradient-boosted decision trees (GBDTs), not the IBU prior-iteration coupling. Comparing the two on the same data quantifies the method-dependence of the result (§3).

Goal and scope. The goal is a faithful, fully uncertainty-quantified reproduction of the published 2D cross section as the demonstrator for unbinned OmniFold on MINERvA data, followed by the available-energy higher-dimensional extension. This note covers the method (§2), the 2D result and its comparison to the published data and to GENIE MINERvA Tune v1 (§3), the uncertainty budget (§4), the validation suite (§5), and the resolution of the open questions raised during review (§7).

2 Method and analysis pipeline

Unbinned OmniFold. OmniFold [3, 4] alternates two reweighting steps per iteration: step 1 reweights the reconstructed simulation to the data in observable space (a learned data/MC likelihood ratio, pulled back to truth through the detector response), and step 2 reweights truth-level simulation to the step-1 result, yielding a self-consistent truth-level weight per event. We use gradient-boosted decision trees as the classifiers/regressors. The engine is `unbinned_unfolding/python/omnifold.py`; the MINERvA driver is `2d-unfolding/unfold_2d_omnifold_unbinned.py` with backends {`exact,hist,lgbm`} (§3 compares them). The unfolded observables are $(p_T, p_{\|})$ of the muon; the phase space is $\theta_\mu < 20°$, $p_T < 4.5\,\mathrm{GeV/c}$, $1.5 < p_{\|} < 60\,\mathrm{GeV/c}$.

Production pipeline (Phase 18.2). The C++ event loop on the MINERvA detector [15] with the NuMI flux of Ref. [16] (`MINERvA101/MINERvA-101-Cross-Section/runEventLoopOmniFold.cpp`) writes the per-event truth and reconstructed trees consumed by the driver. Two invariants make the result truth-tree authoritative:

- **Truth-tree-authoritative reco gate**: a reconstructed signal event is filled only when its (run, subrun, event) key is present in the truth denominator; truth-only events enter as native OmniFold "miss" entries.
- **Completeness** $c =$ (events in the OmniFold input)/(all truth events) is 1.000000 by construction at ME FHC (32 849 103 entries), so the cross-section completeness division is a no-op self-check.

Cross-section extraction. Unfolded truth-level counts are converted to the differential cross section via $d^2\sigma = U/(c\,\Phi\,N\,\mathrm{POT}\,\Delta p_T\,\Delta p_{\|}) \times 10^4$, where Φ is the per-p_T flux integral (broadcast over $p_{\|}$), N the fiducial nucleon count, and POT the data exposure (`extract_cross_section_2d`). The flux normalisation enters as $\sigma \propto 1/\Phi$; correctly varying Φ per flux universe was the dominant systematic fix (§4).

Binning. The reported binning ($14\ p_T \times 16\ p_{\|}$ bins; 205 of 224 reported) follows `minerva_paper_anc/bin_mappin` the 19 unreported diagonal cells have $p_T^{\mathrm{hi}}/p_{\|}^{\mathrm{lo}} > \tan 20°$. The full edge table and reported-bin mapping are recorded in `2D_OMNIFOLD_REFERENCE.md` and in `uq_statistical_methods.tex`.

3 Results

3.1 Double-differential cross section

Figure 1 shows the unfolded $d^2\sigma/(dp_T\,dp_{\|})$ and its 1D projections. The total cross section is $3.073 \times 10^{-38}\,\mathrm{cm}^2/\mathrm{nucleon}$ (published: $3.039 \times 10^{-38}\,\mathrm{cm}^2/\mathrm{nucleon}$).

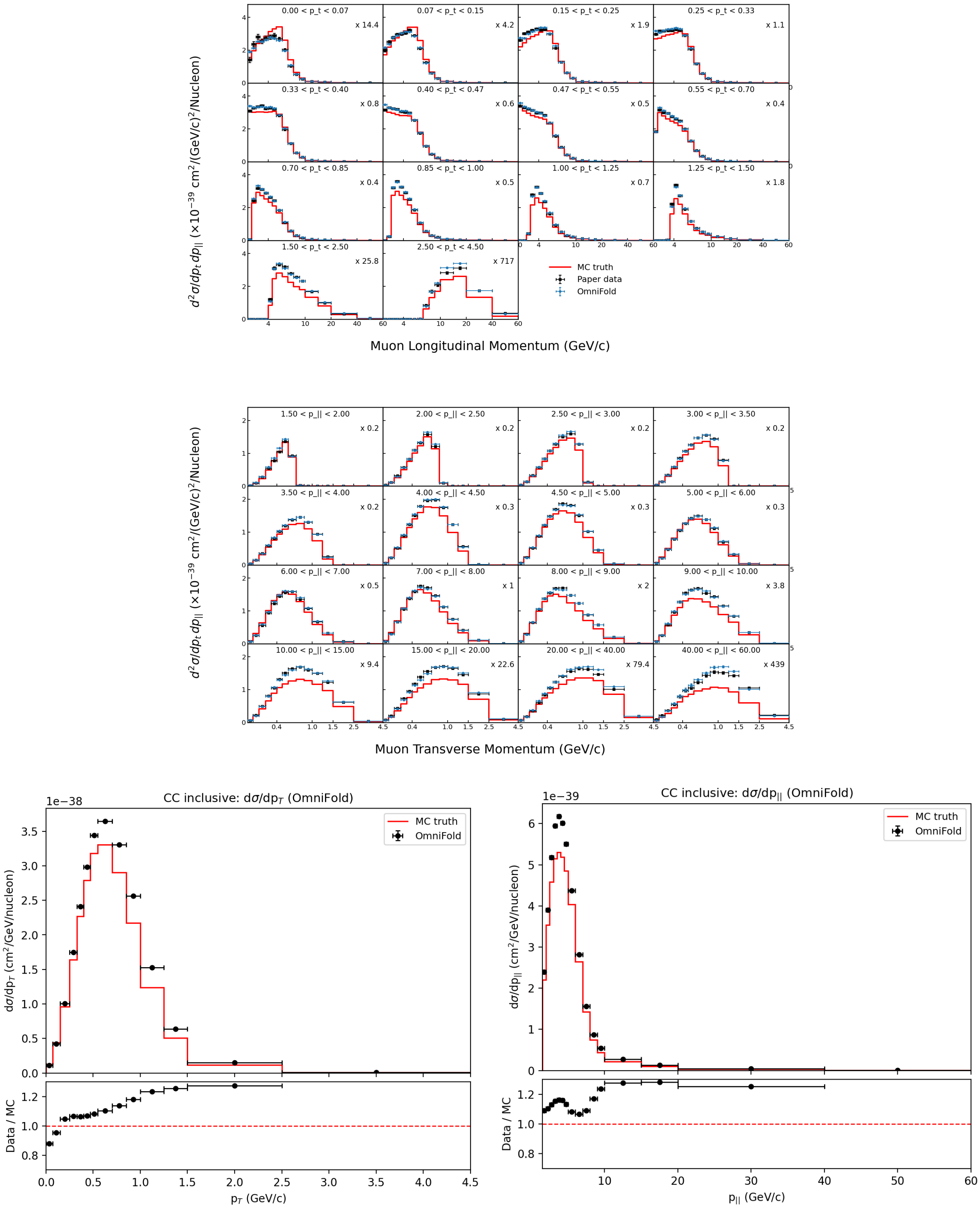

Figure 1: Unfolded 2D cross section (top, Fig. 13 style) and its $p_T/p_\parallel$ projections (bottom).

Quantity (205 reported bins)	Value
σ_{tot}(ours)/σ_{tot}(paper)	1.011
Median bin ratio (ours/paper)	1.006
Bins within 5 %/10 %/20 %	77.6/94.1/98.5 %
χ^2/ndf, paper covariance	3.661
χ^2/ndf, combined covariance (paper + ours)	1.481
Pull mean / RMS (paper covariance)	0.089/0.598

Table 1: Agreement with arXiv:2106.16210. Per-bin pulls (Fig. 2) are sub-1σ.

3.2 Comparison to the published measurement

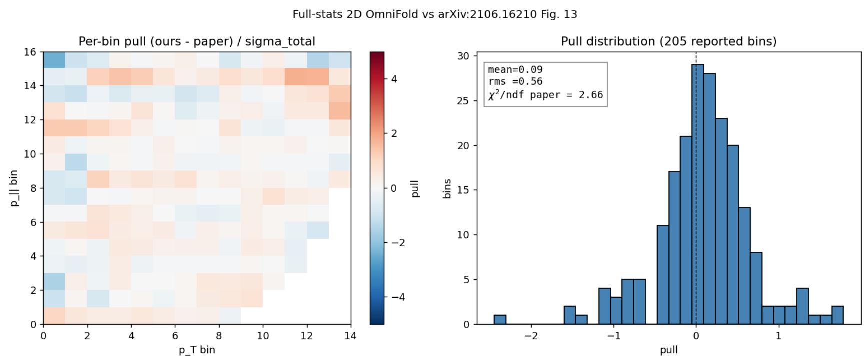

Figure 2: Per-bin pull (ours − paper)/σ on the 14×16 grid and its distribution.

3.3 Anatomy of the $\chi^2 = 3.66$ tension

The central values agree (pull RMS 0.598, so a diagonal-only χ^2/ndf would be 0.37), yet the full paper-covariance χ^2/ndf is 3.66: a 10× inflation living entirely in the off-diagonal structure of the published covariance. The eigenmode decomposition (Fig. 3) shows the tension is carried by moderate-λ modes with 5–6σ eigen-pulls (the 10 smallest-λ modes carry only 3 %; keeping the 73 best-measured directions still gives 1.42), so it is *not* a small-λ inversion artifact. It is *shape*, not normalisation ($\chi^2_{\text{norm}} = 0.10$ vs $\chi^2_{\text{shape}} = 750$), localised to the low-$p_T$ cross-section peak ridge where the flux and muon-energy bands dominate. A $\sim$ 1-unit methodological band comes from the GBDT estimator (exact-GBT 3.66 vs HistGBT 2.70 $\pm$ 0.04 vs LightGBM 2.65; iteration count is negligible, 5 → 10 iters move it by +0.01). Full treatment: `uq_statistical_methods.tex §sec:tension`.

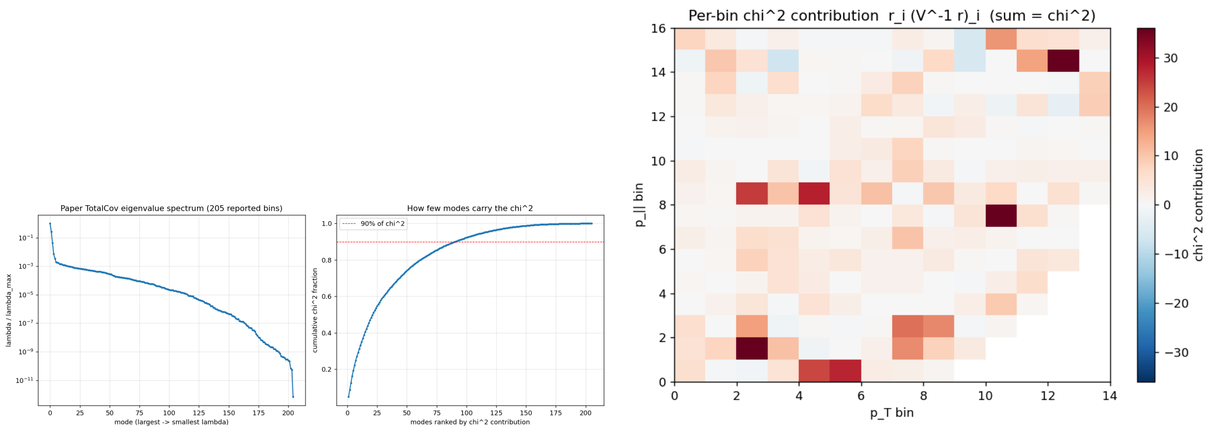

Figure 3: Tension anatomy: eigenvalue spectrum with cumulative χ^2 (left) and the per-bin χ^2-contribution map (right).

3.4 Comparison to GENIE MINERvA Tune v1

Both the published data and our unfolded result are compared to the GENIE 2.12.6 [17] MINERvA Tune v1 prediction shipped with the ancillary (Fig. 4). The tune under-predicts the total by $\sim 9\,\%$ ($\sigma_{\mathrm{tune}}/\sigma_{\mathrm{data}} = 0.91$), concentrated in the low-p_T peak, and is strongly disfavoured by the data covariance (χ^2/ndf: data vs tune 33.0, ours vs tune 26.5). Our unfolded result tracks the data far more closely than the model does. (`compare_to_models.py`.) The published analysis [1] additionally compares to NuWro [18], GiBUU [19] and NEUT [20]; because OmniFold delivers a per-event reweighted truth ensemble, any such generator prediction can be compared to our result in *any* projection or dimension without re-unfolding — a capability we exploit for the 3D extension (§6).

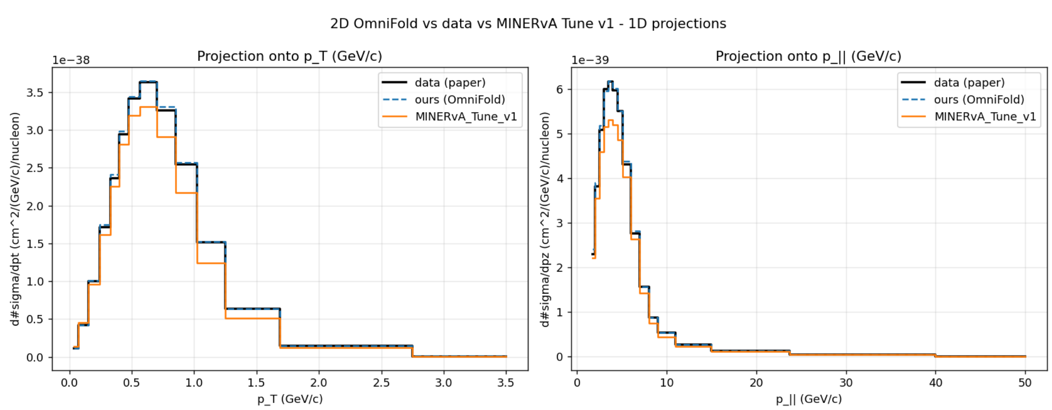

Figure 4: Our result (dashed) on top of the published data (black); MINERvA Tune v1 (orange) dips below in the peak.

4 Uncertainty quantification

The uncertainty budget combines three independent blocks, summed assuming independence (distinct RNGs / physics sources). Full procedures are in `uq_statistical_methods.tex` (App. A); the headline is that our *standalone* combined budget matches the published total.

Component	N	$\sqrt{\mathrm{Tr}\, C}$	per-bin median rel. σ
ML noise (lgbm seedscan)	10	5.06×10^{-41}	0.166 %
Statistical (Poisson bootstrap)	300	1.82×10^{-40}	0.549 %
Systematic (universe sweep + 1.4 % norm)	187+1	3.21×10^{-39}	6.830 %
Combined (block sum)	—	3.22×10^{-39}	6.865 %
Paper TotalCov (reference)	—	2.68×10^{-39}	6.86 %

Table 2: Stage-2 UQ envelope on the 205 reported bins. The standalone combined budget (6.865 %) matches the published total (6.86 %).

Systematic universes. 44 bands (6 lateral, 38 vertical), 187 universes, reweighted/re-run via the MAT-MINERvA universe machinery; the covariance is built MAT-conformant (`MnvVertErrorBand::CalcCovMx`; verified byte-for-byte against MAT `MnvH1D::GetTotalErrorMatrix`, max rel. diff $\sim 10^{-17}$). Top bands: Flux 4.99 %, Muon_Energy_MINOS 2.31 %, Muon_Energy_MINERvA 1.29 %, MinosEfficiency 1.47 %.

Flux $1/\Phi$ normalisation. The dominant flux term enters as $\sigma \propto 1/\Phi$; varying Φ per flux universe (rather than dividing every universe by the CV Φ) raised the Flux band from ~ 1 % to 4.99 % and brought the standalone combined budget to 6.87 %. The 1.4 % target-nucleon normalisation is a fully-correlated rank-1 band.

χ^2 **caveat.** The combined paper+ours $\chi^2/\mathrm{ndf} = 1.481$ double-counts systematics shared with the paper (flux/GENIE/MINOS); the ours-only inverse-covariance χ^2 is evaluated only with a truncated-spectral convention on the well-constrained subspace. We therefore quote per-bin pulls, the regularisation scan, and the truncated result together rather than treating an unconstrained pseudo-inverse as a standalone goodness-of-fit (§7).

5 Validation

Closure. Three closure families, all passing (1A lgbm 5-iter; thresholds in parentheses):

- **Truth-reweight** (multiply truth and reco weights by the same factor): gaussian-p_T residual 0.046 %, tilt-$p_{\|}$ 0.013 % (thr 1.5 %) — injected truth recovered.
- **Hidden-variable** (bias on $\Delta p_T = p_T^{\mathrm{reco}} - p_T^{\mathrm{truth}}$, a resolution variable not in the feature set): leakage linear in amplitude, $A = 0.05/0.10/0.30 \rightarrow 0.57/1.14/3.43$ % (thr 3.0 %).
- **Alt-model** (train on CV, pseudo-data from an alternative universe): MaCCQE 0.068 %, Flux 0.376 % (thr 2.0 %).

Bottom-line test. Following the unbinned-unfolding guide [21], the unfolded result must recover the data's deviation from the prior with a residual far smaller than the feature itself (the unfolding cannot manufacture discriminating power). Using the closure runs, in the bins carrying the injected feature the unfolded result tracks the injected truth to $\approx 10\times$ better than the feature size: 2D (gaussian-bump truth reweight) residual 1.69 % vs injected 17.2 % (ratio 0.098); 3D (+30 % E_{avail}

bump, E_{avail} projection) 1.84 % vs 18.1 % (ratio 0.102). Both pass. (The naive data-vs-prior form—reco vs truth χ^2—instead has truth>reco, as expected for a high-resolution measurement where detector smearing lowers the reco baseline and a stat-only diagonal omits the unfolding covariance; the proper full-covariance goodness-of-fit is the χ^2-vs-paper of §3.3 and the generator comparisons.)

Unbinned goodness-of-fit. As a binning-free check that the unfold actually removes the data $\leftrightarrow$ MC mismatch, we use a classifier two-sample test (C2ST): a classifier is trained to separate the data from the (reweighted) simulation in the full unbinned feature space, and its test-set accuracy is compared to the chance value 0.5. *Before* unfolding the two are separable — accuracy 0.523, AUC 0.535, $z = 33.4$ ($p \approx 5 \times 10^{-244}$); *after* unfolding the classifier is driven to chance — accuracy 0.501, AUC 0.501, $z = 1.4$, $p = 0.17$. Data and unfolded simulation are thus statistically indistinguishable with no detectable residual structure: the unbinned analogue of a binned goodness-of-fit, and it passes.

Coverage. The UQ note records a 200-toy closure+bootstrap campaign: mean coverage 68.71 % vs Gaussian 68.27 %; mean |residual|/σ 0.794 vs $\sqrt{2/\pi} = 0.798$; and 97.6 % of bins meeting the 65 % target. The current checkout retains only the 20-toy stage-1 summary under `uq/coverage/`; restore the 200 toy ROOTs before independently regenerating this number from disk.

Stability. 5-iter total σ within 0.026 % of 10-iter (per-bin shape RMS 1.54 %); HistGBT and exact-GBT agree on total σ to 0.04 %. Efficiency (Fig. 6) reproduces the published Fig. 5.

Classifier choice (GBDT vs. NN). Published OmniFold analyses use neural-network classifiers [3, 21]; we use gradient-boosted decision trees, whose output must still be a calibrated posterior for $w = p/(1-p)$ to be valid. On the step-1 reco classifier (data vs. MC reco) the reliability curve is near-diagonal (Brier ≈ 0.25 at AUC ≈ 0.537: data and reco MC are nearly indistinguishable, so the reweight is a modest correction). Across the $(p_T, p_{\|})$ bins the true data/MC ratio still spans ≈ 100 %; the GBDT recovers it to 4.7 % median, at least as well as a dense MLP (20.9 %, untuned), and the two reweights agree with bin-wise correlation 0.92 — the learned reweight is robust to the classifier family. A full scalar keras-MLP OmniFold cross-check on the 3D inputs gives total ratio 1.0078 versus the GBDT reference and median projection differences of 0.7 % to 1.4 %, consistent with the assigned ML band. As a stronger family cross-check, a vendored PET (point-cloud transformer [22]) unfold was run on the actual per-hadron point clouds — the non-muon reconstructed clusters (`cluster_energy`, `cluster_pos`, `cluster_z`, filtered on `cluster_isMuontrack`) read by `CVUniverse::GetRecoClusters`, after an earlier attempt that used the wrong, mostly-empty `ExtraEnergyClusters_*` branch was discarded. Area-normalized against the frozen 4D GBDT result, the PET unfold reproduces the scalar shape to a median 2.3 % to 3.9 % per bin (p_T 3.9 %, $p_{\|}$ 2.4 %, E_{avail} 2.6 %, q_3 2.3 %; Fig. 5). This is a shape/method-robustness check on a 2M-event subsample, not an independent physics result: it shows the learned reweight is stable across both the classifier family (GBDT vs. MLP vs. transformer) and the input representation (engineered scalars vs. raw point cloud).

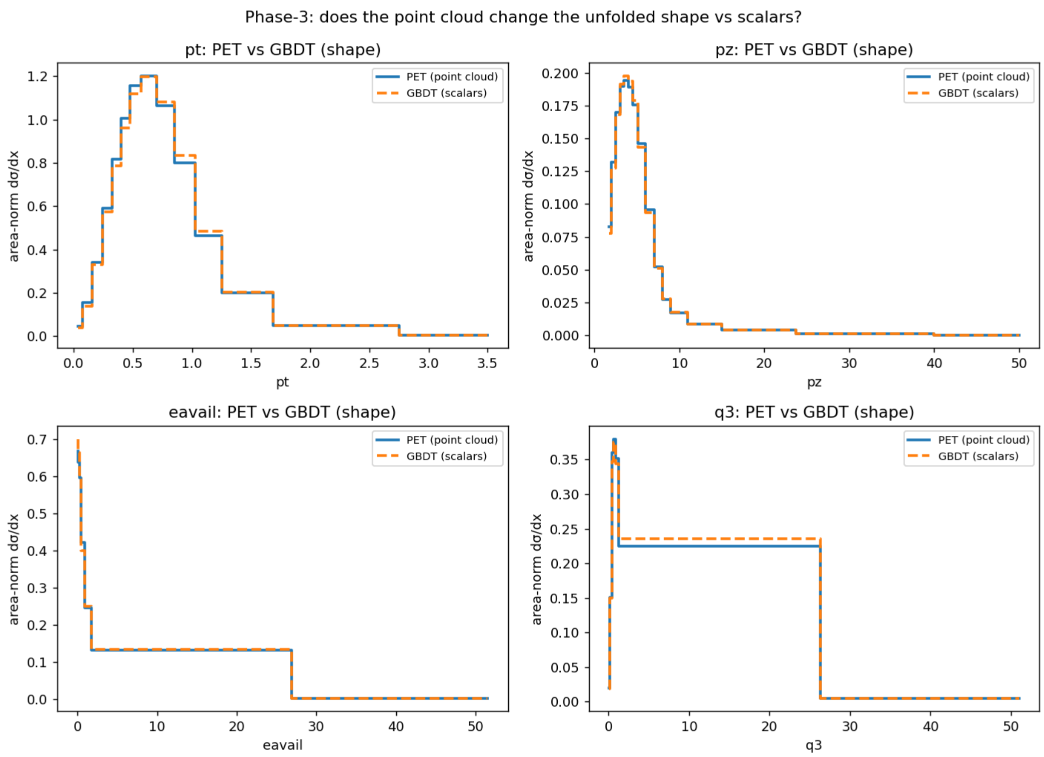

Figure 5: Method-robustness check: area-normalized 4D unfolded shapes from the PET point-cloud transformer (solid) versus the production scalar GBDT (dashed), projected onto each axis. Median per-bin shape differences are 2.3 % to 3.9 %; the only visible deviation is the wide q_3 catch bin, a subsample-statistics effect. Shape-only comparison (PET on a 2M-event subsample), not an independent physics result.

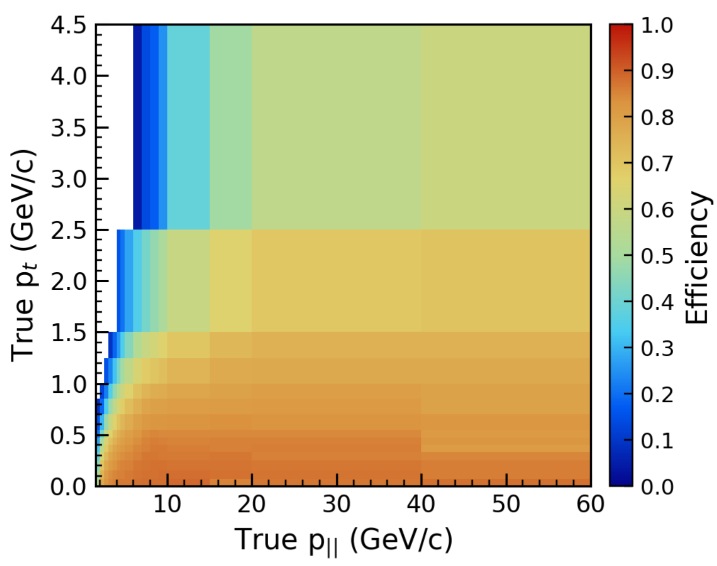

Figure 6: Selection efficiency (Fig. 5 style).

6 Higher-dimensional extension: available energy

The enabling property of unbinned OmniFold is that adding an observable is adding an input feature: the step-1/step-2 classifiers simply gain a column, with no migration-matrix or binning penalty. There is no practical D'Agostini IBU analogue, which makes this the primary physics motivation for the technique. We exercise it by extending the measurement to $d^3\sigma/(dp_T\,dp_\parallel\,dE_{\rm avail})$ with available energy as a third axis.

The significance is clear from the measurement record: every published MINERvA differential cross section has unfolded at most two kinematic variables simultaneously (Fig. 7, Table 3), the double-differential measurements all relying on binned IBU. A simultaneous *three*-observable unfold of MINERvA data is, to our knowledge, without precedent, and is a direct consequence of the unbinned formulation.

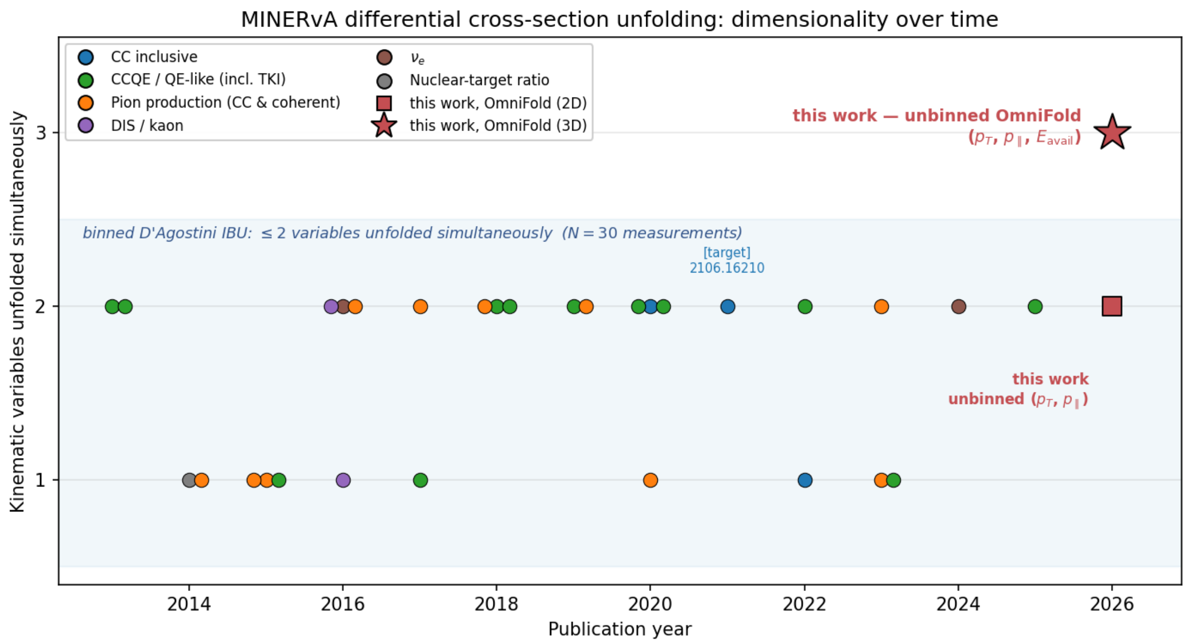

Figure 7: Dimensionality of the full set of MINERvA differential cross-section measurements over time (number of kinematic variables unfolded *simultaneously*; a double-differential $d^2\sigma/dx\,dy$ counts as two), coloured by channel. All $N = 30$ prior measurements use binned D'Agostini IBU and reach at most two variables; this work's unbinned OmniFold adds a third. Source: the MINERvA data-release page; full list in Table 3. Dimensionality is classified from each analysis's headline differential result; flux, reconstruction, generator-tuning and total-only entries are excluded.

Table 3: Full MINERvA differential cross-section survey: the number of kinematic variables unfolded simultaneously (Dim.), classified from each analysis's headline differential result. Channels: CC-inclusive (incl.), CCQE/QE-like incl. transverse kinematic imbalance (QE), pion production CC and coherent (π), deep-inelastic / kaon (DIS), electron-neutrino (ν_e), nuclear-target ratio (ratio).

Year	Channel; observables	Dim.	Ref.
2013	QE; ν_μ CCQE $p_T, p_\parallel$	2	[23]
2013	QE; $\bar{\nu}_\mu$ CCQE $p_T, p_\parallel$	2	[24]
2014	ratio; CC on C/Fe/Pb to CH	1	[25]
2014	π; coherent $\pi^\pm$ (total & diff.)	1	[26]
2015	π; CC $1\pi^+$ (T_π, θ_π)	1	[27]
2015	QE; μ+p final states	1	[28]
2015	π; $\bar{\nu}_\mu$ single π^0	1	[29]
2016	ν_e; CCQE-like on CH	2	[30]
2016	π; $\nu_\mu/\bar{\nu}_\mu$ pion prod.	2	[31]
2016	DIS; partonic nuclear effects	2	[32]
2016	DIS; CC K^+ production	1	[33]
2017	π; CC single π^0	2	[34]
2017	QE; nuclear-dependence (multi-target)	1	[35]
2018	QE; $\bar{\nu}_\mu$ QE-like $p_T, p_\parallel$	2	[36]
2018	QE; μ-p mesonless TKI	2	[37]
2018	π; coherent $\pi^\pm$ on C	2	[38]
2019	QE; quasielastic-like ν_μ	2	[39]
2019	π; $\bar{\nu}_\mu$ single π^-	2	[40]
2020	incl.; CC-incl $\langle E_\nu \rangle \sim 3.5$ GeV $p_T, p_\parallel$	2	[41]
2020	QE; nuclear binding & TKI	2	[42]
2020	QE; high-statistics QE-like	2	[43]
2020	π; CC π^0 nuclear effects	1	[44]
2021	incl.; CC-incl $\langle E_\nu \rangle \sim 6$ GeV $p_T, p_\parallel$ **(target)**	2	[1]
2022	QE; QE-like muon & proton kinematics	2	[45]
2022	incl.; CC-incl low momentum transfer	1	[46]
2023	π; CC $1\pi^+$ (multi-target)	2	[47]
2023	π; coherent π^+ (multi-target)	1	[48]
2023	QE; $\bar{\nu}_\mu$ multi-neutron low-E_{avail}	1	[49]
2024	ν_e; low-Q^2 (E_{avail}, p_T)	2	[50]
2025	QE; A-dependence TKI (multi-target)	2	[51]
2026	**this work**; unbinned $p_T, p_\parallel$	2	—
2026	**this work**; unbinned $p_T, p_\parallel, E_{\text{avail}}$	**3**	—
2026	**this work**; unbinned $p_T, p_\parallel, E_{\text{avail}}, q_3$	**4**	—

6.1 Definitions and pipeline

Available energy follows the MINERvA low-recoil convention (Ref. [50], Eq. 4): truth $E_{\text{avail}} = \sum_p \text{KE} + \sum_{\pi^\pm} \text{KE} + \sum_{\pi^0,\gamma} E$ (excluding neutrons and the muon), via `CVUniverse::GetEAvailableTrue()`; reco E_{avail} is the calorimetric tracker+ECAL estimate scaled by 1.17, `CVUniverse::NewEavail()`. Both accessors were added to the event loop, which was re-run over all twelve ME FHC playlists to append the E_{avail} branches alongside the existing $p_T/p_\parallel$ branches. The unfolding driver is the 2D pipeline with a three-column feature stack ($p_T, p_\parallel, E_{\text{avail}}$); the cross-section extraction generalises to a 3D Jacobian $\Delta p_T \, \Delta p_\parallel \, \Delta E_{\text{avail}}$ with the flux Φ still applied per p_T (broadcast over $p_\parallel$ and

E_{avail}). The E_{avail} binning $[0, 0.1, 0.2, 0.4, 0.8, 1.5, 3.0, 100]$ GeV ends in a wide catch bin so that the E_{avail}-marginal captures the full CC-inclusive recoil tail; the global OmniFold completeness is $c = 1.0000$, as in 2D.

6.2 The available-energy spectrum

Figure 8 shows the unfolded $d\sigma/dE_{\mathrm{avail}}$ (the 1D projection of the 3D result). It falls monotonically from 2.19×10^{-38} cm^2/GeV/nucleon at the low-recoil peak to 6.8×10^{-41} in the catch bin — the expected CC-inclusive shape, with most of the rate at $E_{\mathrm{avail}} \lesssim 0.8$ GeV.

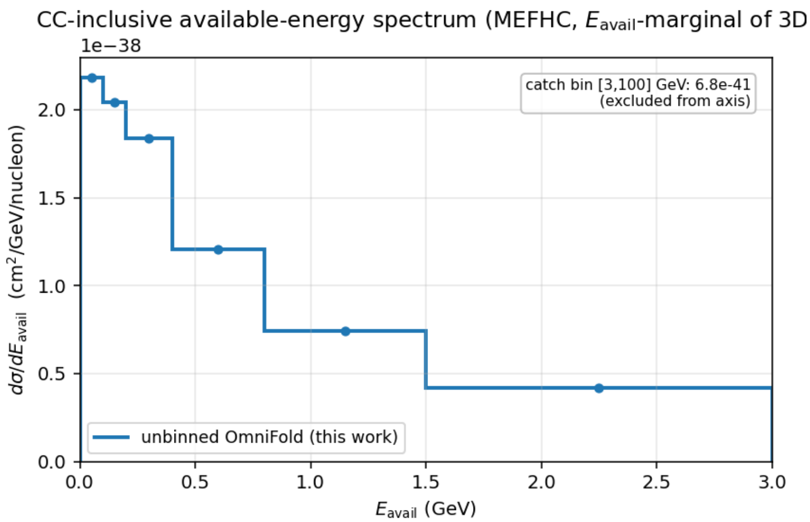

Figure 8: Unfolded available-energy spectrum $d\sigma/dE_{\mathrm{avail}}$, the 1D projection of the 3D measurement. The wide $[3, 100]$ GeV catch bin (value annotated) is excluded from the axis.

6.3 Validation against the 2D measurement

With no published 3D reference, the validation anchor is that marginalising the 3D cross section over E_{avail} must reproduce the 2D result of §3. The marginal recovers the 2D *normalisation* closely: total σ agrees to $+0.95\,\%$ and the per-bin marginal/2D ratio has median 1.0016. The *shape* agreement is looser — the marginal versus the published covariance gives $\chi^2/\mathrm{ndf} = 4.98$, above the frozen 2D values (3.66 default, 2.65 LightGBM-CV), driven by a $\sim 4.4\,\%$ per-bin scatter that the third axis injects into the $(p_T, p_{||})$ marginal (Fig. 9).

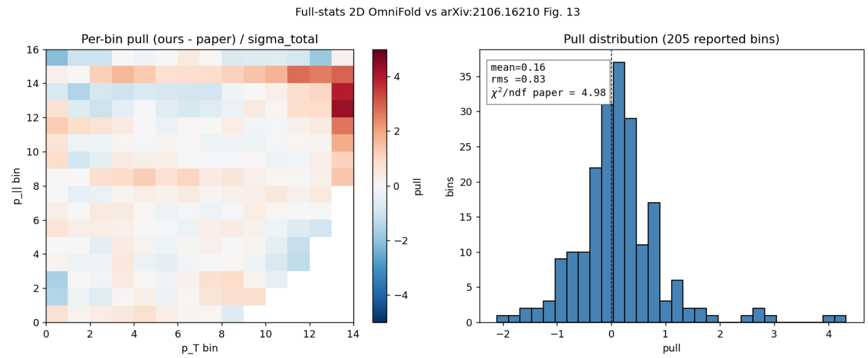

Figure 9: Per-bin pull of the E_{avail}-marginal against the published covariance, on the 14×16 grid and its distribution.

6.4 Closure: the scatter is physical, not a method bias

A reweight closure isolates whether that 4.4 % scatter is method bias or real information. We inject a known +30 % Gaussian bump in *truth* E_{avail} (centre 0.3 GeV, width 0.15 GeV) into the pseudo-data and require the 3D unfold to recover it. It does: the recovered step-2 mean weight is 1.048 against the injected 1.0485, and the residual of the unfold against the reweighted-truth reference is, on the E_{avail}-marginal, median 0.9999 with standard deviation 0.0006 (0.06 %); the per-3D-bin residual standard deviation is 3.2 % (MC statistics in sparse high-recoil corners). The closure marginal scatter is thus $\sim 70\times$ tighter than the 4.4 % data-versus-2D scatter. The method is therefore unbiased on the new axis, and we read the elevated marginal χ^2 as *genuine data ↔ MC structure that the available-energy dimension exposes* — information that is invisible to the 2D measurement — rather than a pipeline artefact (the total normalisation matches to < 1 %).

This establishes the 3D framework and its first closure.

6.5 Generator comparison in 3D

Because the published ancillary ships only binned 2D model histograms, a quantitative comparison of the 3D result (or of arbitrary projections) requires generating truth-level events ourselves. No detector simulation is needed — the unfolded result already lives at truth level — so a generator prediction is obtained by histogramming its truth $d^3\sigma$ in the same fiducial phase space and the same truth E_{avail} definition. We compare four predictions:

- **GENIE 2.12.10 central value** — generated with `gevgen` on a CH target with the MINERvA ME FHC flux and the DefaultPlusValenciaMEC configuration [17].
- **MINERvA Tune v1** — the analysis tune (base GENIE + RPA, low-recoil 2p2h, and non-resonant-pion reweights), extracted directly from the event-loop MC truth (the `w_truth` weights), which reproduces the shipped ancillary curve to 0.01 % in normalisation.
- **NuWro 21.09** [18] — an independent generator (C target), the same flux and phase space.
- **GiBUU 2019** [19] — a transport-theory generator (C target), built and run *natively* on the compute system (no container), same flux and phase space.

NEUT [20] is not included (no accessible build at the time of writing). Each prediction is normalised by its own flux-averaged total CC cross section per nucleon ($\langle\sigma\rangle \approx 3.6$–$4.0 \times 10^{-38}\,\text{cm}^2$).

Figure 10 overlays the four on the unfolded result, with the combined-covariance systematic band of §6.6 on the data. The flux-averaged totals in the measurement phase space order as GiBUU $(2.22\times10^{-38}) <$ NuWro $(2.34\times10^{-38}) <$ GENIE CV $(2.52\times10^{-38}) <$ Tune v1 $(2.71\times10^{-38}) <$ data $(3.08\times10^{-38}\,\mathrm{cm}^2/\mathrm{nucleon})$: all four models under-predict the rate, by 12 % to 28 %. On $(p_T, p_\parallel)$ the four track the data shape closely; the discriminating axis is E_{avail}, where NuWro, Tune v1 and GiBUU *suppress* the low-recoil quasielastic peak below bare GENIE (RPA and nuclear effects) — yet the data sit *above* all four in the lowest-E_{avail} bins. This low-available-energy excess is invisible to the 2D measurement and is a direct statement of the unbinned method's high-dimensional reach.

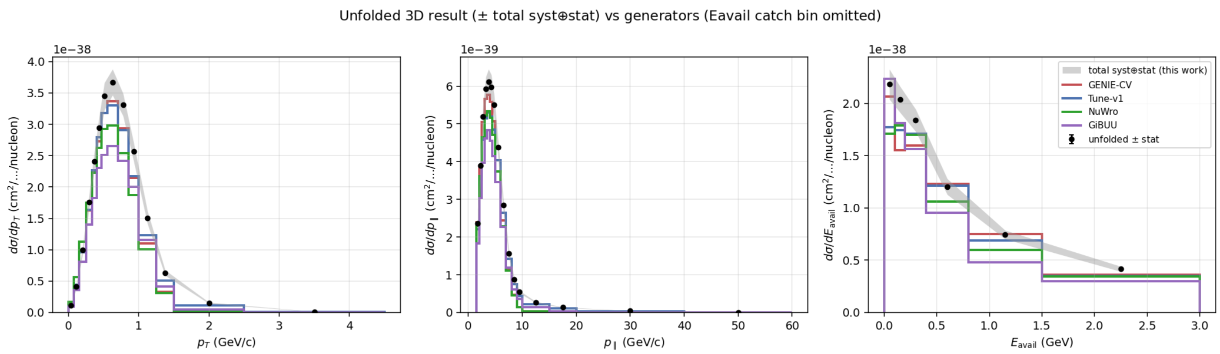

Figure 10: 1D projections of the unfolded 3D cross section (black, with the combined-covariance systematic band) versus four generator predictions produced at truth level: GENIE 2.12 CV, MINERvA Tune v1, NuWro 21.09, and GiBUU 2019. On p_T and $p_\parallel$ the models track the data shape; on E_{avail} they split, and all under-predict the data at low available energy. The wide E_{avail} catch bin is omitted from the axis.

Quantitatively, against the combined 3D covariance (§6.6) the integrated E_{avail} rate (catch bin dropped) places the models at $-7.2\,\%$ (GENIE CV, 1.3σ), $-9.5\,\%$ (Tune v1, 1.8σ), $-15.3\,\%$ (NuWro, 2.9σ) and $-21.9\,\%$ (GiBUU, 4.1σ) below the data. A full-3D goodness-of-fit over all 1431 bins, evaluated with a truncated-spectral χ^2 on the rank-247 constrained subspace of the combined covariance (a raw pseudo-inverse of the rank-deficient matrix is never taken), excludes all four at $p \approx 0$ but ranks them sharply: diagonal-only χ^2/ndf is Tune v1 **4.8**, GiBUU 32.4, GENIE CV 34, NuWro 36; the full correlated metric is Tune v1 131, NuWro 1287, GENIE CV 1512, with GiBUU worst of all because 23.5 % of its residual lies *outside* the constrained subspace (its 3D shape differs in directions the covariance barely constrains) on top of the largest normalisation offset.

6.6 The 3D systematic-uncertainty campaign

The band on Fig. 10 and the χ^2 above rest on a full 3D analogue of the 2D uncertainty budget of §4. We re-ran the event loop with the systematic universes dumped on the three-axis $(p_T, p_\parallel, E_{\mathrm{avail}})$ grid (a 187-universe sweep), propagated the bootstrap statistical and ML-seedscan covariances, and summed them into a single combined covariance $C_{\mathrm{syst}} + C_{\mathrm{stat}} + C_{\mathrm{ML}}$ over all 1431 reported bins. The result has $\sqrt{\mathrm{tr}} = 5.72 \times 10^{-39}$, a median per-bin relative uncertainty of 10.4 %, and numerical rank 247 (of 1431); it is flux-dominated, with the same systematic band ordering as the 2D measurement. This combined covariance is what projects onto each axis for the per-bin bands and onto the full space for the goodness-of-fit, and it is the internally-correct reference for the "ours-only" χ^2 given that our bootstrap statistical block is genuinely smaller than the paper's (an OmniFold efficiency, confirmed by the data/MC split; see §7).

6.7 The low-available-energy excess recovers the known low-recoil 2p2h deficit

The low-E_{avail} excess is the most physically interesting feature the third axis exposes, and the 3D framework lets us identify its origin directly by generating the relevant physics ourselves rather than only reweighting. It sits in a well-established MINERvA low-recoil program: the enhancement of the event rate between the quasielastic and Δ peaks, with multi-nucleon final states, was first isolated in the low-three-momentum-transfer subsample [52] and is the effect the empirical MINERvA tune was built to absorb — our measurement re-exposes it on the muon-kinematics $\times\, E_{\text{avail}}$ axes.

Final-state interactions do not explain it. A `grwght1p` reweight pass on the 2M central-value GENIE events varying the pion-FSI dials shows both are sub-percent on $d\sigma/dE_{\text{avail}}$: `FrInel_pi` shifts the spectrum by $\leq 0.74\,\%$ (peak in 0.10–0.20 GeV) and `FrAbs_pi` by $\leq 0.82\,\%$ (peak in the lowest bin), with the integrated rate moving only ± 0.02–$0.03\,\%$ at $\pm 1\sigma$. A $< 1\,\%$ knob cannot bridge a 7–15 % excess, so the excess points to the initial-state/nuclear model, not rescattering.

The shape and size match 2p2h. A subtlety drives the diagnosis: our base GENIE central value was generated with the default event-generator list, which contains *no* meson-exchange-current (2p2h/MEC) events (`mec` $= 0$ for all 1.48×10^6 CC events; the modes are QE 11 %, RES 22 %, DIS 66 %, COH 0.9 %). Any 2p2h is therefore *added*, not reweighted. Decomposing the central-value $d\sigma/dE_{\text{avail}}$ by interaction mode (per-bin mode count-fractions $\times$ the bin value) and overlaying the data localises the excess to the quasielastic–Δ dip: the $[0.10, 0.20)$ bin is $+3.9\sigma$ and $[0.20, 0.40)$ is $+2.3\sigma$, while the QE-dominated $[0, 0.10)$ already matches ($+0.8\sigma$) and $[0.40, 1.50)$ match to $\pm 0.4\sigma$. 57 % of the integrated deficit sits at $E_{\text{avail}} \leq 0.4$ GeV, exactly where 2p2h lives, and closing it requires a 2p2h component $\approx 43\,\%$ of the QE rate (62 % locally) — the standard empirical/Valencia-MEC magnitude, in contrast to the sub-percent FSI dials (Fig. 11).

Enabling Valencia 2p2h confirms it. We regenerated the GENIE central value with the empirical 2p2h turned on (`--event-generator-list Default+CCMEC`, 2M events). Because the Nieves–Simo–Vacas MEC channel, though present in the splines, is absent from the `gspl2root` total-CC graph, the sample is normalised by the non-MEC CC count (anchoring QE+RES+DIS+COH to the known total and letting MEC add on top by $1/(1 - f_{\text{MEC}})$, with $f_{\text{MEC}} = 2.87\,\%$). The added 2p2h lands squarely in the dip and moves GENIE toward the data: the $[0, 0.10)$ bin goes from $+0.8\sigma$ to -0.04σ (onto the data), $[0.10, 0.20)$ from $+3.9\sigma$ to $+2.85\sigma$, and $[0.20, 0.40)$ from $+2.3\sigma$ to $+1.0\sigma$ (Fig. 12). It fills 46 % of the data$-$CV gap in the dip and 27 % of the integrated deficit (CV $-7.2\,\% \to -5.2\,\%$), and improves the full-3D χ^2/ndf from 1512 to 1145 ($-24\,\%$). It remains well short of Tune v1 (131), because Tune v1 layers MINERvA's empirical low-recoil 2p2h *enhancement* and RPA on top of the stock Valencia MEC — i.e. stock Valencia 2p2h is real but under-strength for MINERvA low recoil, which is precisely the known MINERvA-tune story and is consistent with NuWro and GiBUU (native, un-enhanced 2p2h) sitting low as well. The one feature 2p2h does not touch is a separate $+2.2\sigma$ excess in the high-E_{avail} $[1.50, 3.00)$ bin, a DIS-tail modelling issue (§7).

This low-available-energy deficit is the same feature MINERvA's dedicated low-recoil inclusive measurement reports: Ascencio *et al.* [46], which measures $d^2\sigma/(dq_3\, dE_{\text{avail}})$ for $q_3 < 1.2$ GeV at the same $\langle E_\nu \rangle \sim 6$ GeV, finds GENIE and NuWro underpredicting the data in the low-E_{avail} region absent an enhanced 2p2h component. Our excess appears in the muon-kinematics extension (integrated over all q_3) rather than at fixed q_3, so the comparison is of the E_{avail} *shape* and direction rather than absolute normalisation; the consistency confirms that the OmniFold 3D extension recovers the established MINERvA low-recoil/2p2h feature as a by-product of the muon-kinematics unfolding. The overlay tooling (`3d-unfolding/genie/compare_ascencio_eavail.py`) is in place for a quantitative shape comparison once the [46] data release is loaded.

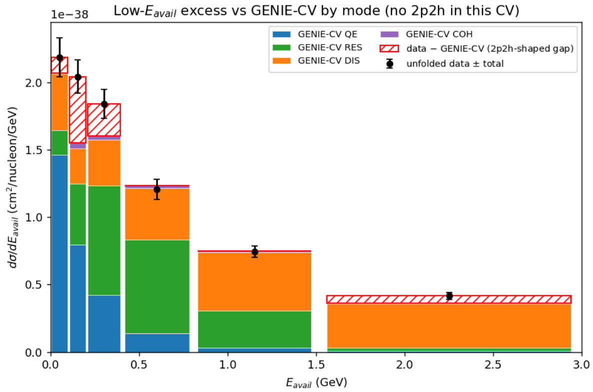

Figure 11: Central-value $d\sigma/dE_{\text{avail}}$ decomposed by interaction mode and overlaid with the unfolded data. The data excess is localised to the quasielastic–Δ dip ($E_{\text{avail}} \leq 0.4\,\text{GeV}$), the shape and size of a missing 2p2h component.

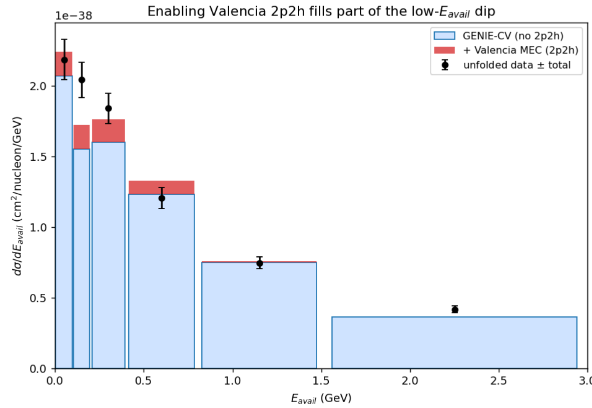

Figure 12: Re-generating GENIE with the empirical Valencia 2p2h enabled (red, stacked on the no-2p2h CV) fills part of the low-E_{avail} dip — about half the data$-$CV gap, the remainder being the MINERvA-tune enhancement.

6.8 Fourth axis: three-momentum transfer q_3

The same one-feature-per-observable property lets us add a fourth axis at no methodological cost. We choose the three-momentum transfer $q_3 = |\vec{q}|$ because (E_{avail}, q_3) is the plane in which the quasielastic, 2p2h and resonance processes separate, and because it makes the comparison to MINERvA's dedicated low-recoil measurement (Ascencio *et al.* [46], $d^2\sigma/dq_3\,dE_{\text{avail}}$) bin-compatible. Truth q_3 is the generator momentum transfer (`CVUniverse::Getq3True()`); reco q_3 is the calorimetric estimator built from the muon Q^2 and the recoil system (`CVUniverse::RecoQ3()`). Both accessors were added and the event loop re-run over all twelve playlists, exactly as for E_{avail}. Unlike E_{avail}, q_3 is *not* invariant under the lateral systematic shifts, so the shifted-q_3 universes are dumped per universe rather than reused from the central value. The driver feature stack becomes $(p_T, p_\|, E_{\text{avail}}, q_3)$; the q_3 binning $[0, 0.2, 0.4, 0.6, 0.8, 1.2, 2.0, 100]$ GeV is deliberately coarse (seven bins) and ends in a catch bin, for $14 \times 16 \times 7 \times 7 = 10\,976$ four-dimensional bins (4830 populated).

Marginal closure. Marginalising the 4D result over q_3 reproduces the 3D E_{avail} spectrum and its low-recoil excess (Fig. 13, right): the E_{avail}-marginal data/GENIE-CV ratio is 1.21 in the lowest bin, dips to 1.01 across the quasielastic–Δ region, and rises again to 1.16–1.23 in the high-E_{avail} DIS tail — the same shape established in 3D (§6.7), now recovered as a by-product of the higher-dimensional unfold (integrated data/GENIE-CV = 1.13). This is the 4D→3D analogue of the 3D→2D Jacobian anchor of §6.3.

What q_3 adds. The new content is the resolved (E_{avail}, q_3) plane (Fig. 13, left). Integrated over E_{avail} the q_3-marginal discrepancy is mild (1.03–1.19, centre panel), but the joint map is structured: at fixed E_{avail} in the quasielastic–Δ dip the data/GENIE-CV ratio *rises* with q_3, the central value over-predicting the low-q_3, moderate-E_{avail} cells (ratio down to 0.66) while the data excess concentrates at low E_{avail} (all q_3) and toward high q_3. This is precisely the correlation a muon-only or fixed-q_3 measurement cannot expose, and it is the natural input for the bin-compatible Ascencio comparison. We show it here at central-value level (the base GENIE event-generator list contains no 2p2h, as in §6.7); per-cell significances await the combined 4D covariance below.

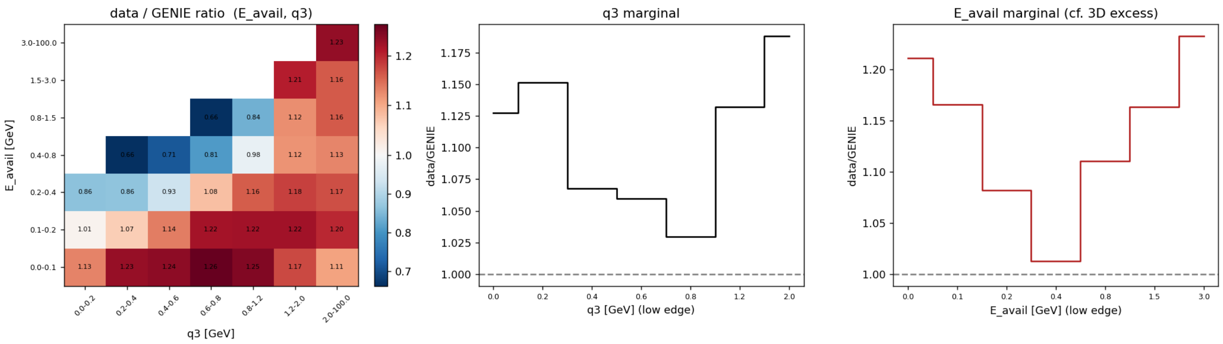

Figure 13: Data/GENIE-CV ratio of the 4D unfold projected onto the (E_{avail}, q_3) plane (left; blank cells are kinematically empty), with the q_3 (centre) and E_{avail} (right) marginals. The E_{avail} marginal reproduces the 3D low-recoil excess; the joint plane shows the data − MC discrepancy to carry genuine q_3 structure (the ratio rising with q_3 across the quasielastic–Δ dip). GENIE central value, no 2p2h in the base event-generator list.

Systematic covariance. A 187-universe systematic sweep on the 4D grid (the same band set as 2D/3D, the shifted-q_3 lateral universes dumped per universe) yields a syst-only covariance with $\sqrt{\text{tr}} = 2.09\times10^{-38}$, median per-bin relative uncertainty 13.4 % (84th percentile 22 %), and numerical rank 142 of 4830 reported bins. The rank is set by the number of independent throws, not the bin count: it is essentially the 3D syst-only value (139), since both budgets use the same ∼40 one-sided bands plus the 100-universe flux multisim. Adding the q_3 axis multiplies the bins (1431 → 4830) without adding throws, so the covariance is simply more rank-deficient (9.7 → 2.9 % of bins spanned) and the truncated-spectral χ^2 of §6.5 becomes correspondingly more essential. The band *ordering* also shifts: where the 2D/3D budgets are flux-dominated, the 4D budget is model-dominated — the 2p2h band is the single largest contributor, followed by the M_A^{RES}, muon-energy-scale and M_A^{CCQE} bands, with flux sub-dominant (∼ 2.8 % median). This is the expected consequence of binning in q_3, the variable that most directly discriminates the interaction models, and it identifies where a q_3-resolved measurement is both most informative and most systematics-limited. The combined budget $C_{\text{syst}} + C_{\text{stat}} + C_{\text{ML}}$ (folding in the 4D bootstrap and ML-seedscan blocks, which raise the rank into the ∼250 range as in 3D) is complete (median per-bin 13.5 %).

Unified-throw cross-check and adoption. The block sum above adds each systematic band's covariance independently; it omits the nonlinear cross-band term that arises because the OmniFold reweighter is not linear in the input perturbations. We quantify that term with a *unified throw*: 160 joint re-unfolds in which all 12 reweightable knob bands and the flux multisim are perturbed simultaneously per event ($w_{\text{cv}} \prod_b \rho_b^{g_b}$, $g_b \sim \mathcal{N}(0,1)$) and the cross section re-extracted, with the OmniFold run-to-run jitter floor subtracted via a zero-shift null set. On the real $(p_T, p_\parallel, E_{\text{avail}}, q_3)$ binning the jitter-corrected unified covariance is **2.0**× the block sum in $\sqrt{\text{tr}}$ — i.e. the block sum *underestimates* the vertical systematic by a factor of two. The inflation is concentrated rather than diffuse (the per-bin σ median ratio is 1.00, but the top 1 % of bins by variance-excess carry 78 % of the trace increase), and those bins are the high-p_T, lowest-E_{avail} (< 0.1 GeV) corner where the migration matrix is most off-diagonal and bands couple most strongly. Because 160 throws cannot estimate a 4830 × 4830 matrix (a direct swap is badly rank-deficient and non-PSD), we *adopt* the result by transferring the throw's per-bin fractional inflation $g_i = \max(\sigma_i^{\text{uni}}, \sigma_i^{\text{blk}})/\sigma_i^{\text{blk}} \geq 1$ onto the sweep's own vertical block, $C \to (C - C^{\text{vert}}) + G\, C^{\text{vert}} G$, which is positive-semidefinite by construction and conservative (never below the block baseline). The adopted combined 4D covariance carries $\sqrt{\mathrm{tr}} = 3.85 \times 10^{-38}$ (median per-bin 13.5 % to 14.9 %). This refines the coarse 3D probe (1.16 median / 1.40 trace) and is the published 4D systematic object.

7 Resolution of the open questions

The earlier draft listed six items for collaboration input. The 2026-06-03 audit (scripts under `2d-unfolding/uq/` and `3d-unfolding/genie/`; full log in `LITERATURE_NOTES.md`) closes or reduces all of them. Each item below states its resolution and what, if anything, still needs a collaborator. Technical detail is in `uq_statistical_methods.tex §sec:rank`.

1. **Flux ↔ Muon_Energy joint block and the rank gap — *resolved.*** The premise of the original question ("the paper sums more bands than we propagate") was wrong. Our systematic-only covariance is rank 140/205; the paper's (`TotalCov` − `StatOnly`) is rank **120** — *lower* — so we propagate at least as many independent systematic directions, using the same `GetStandardSystematics`. The paper's V is full-rank only because it includes the *statistical* block; we match that with $\mathbf{C}^{\mathrm{syst}} + \mathbf{C}^{\mathrm{boot}}$ (rank 201). The one named cross-band correlation, the Bashyal flux ↔ muon-energy-scale block [53], *is* rederivable and now built (`uq/rederive_flux_muonE_cross.py` → `uq/flux_muonE_cross.root`): the muon-energy-scale systematic is rank-1, so the cross term is rank-2, with $\rho = 0.900$ (two independent estimators, 0.87–0.90). It is physically real but adding it leaves the rank unchanged (140 → 140) — its directions already lie in range($\mathbf{C}^{\mathrm{syst}}$). The residual stat-block difference (our bootstrap 2.5× smaller by trace) is explained by the data/MC split bootstrap: it is present in the *data-only* stream ($\mathbf{C}_{\mathrm{data}}$/`StatOnly` = 0.356), i.e. genuine unbinned-OmniFold statistical efficiency, not a missing MC term. *Remaining (optional):* the fitted $(\Phi_{\mathrm{norm}}, \Phi_{\mathrm{shape}}, \mathrm{MuonE}_{\mathrm{scale}})$ posterior from MAT-MINERvA would give a fully paper-independent ρ.

2. **FrInel_pi exclusion — *resolved (negligible).*** An independent `grwght1p` pass on the 2M CV events shows both pion-FSI dials are sub-percent on $d\sigma/dE_{\mathrm{avail}}$: `FrInel_pi` $\leq 0.74\,\%$ and `FrAbs_pi` $\leq 0.82\,\%$ (§6.7). A $< 1\,\%$ knob cannot bridge the 7–15 % low-E_{avail} excess, and we are consistent with MAT in excluding it. *Remaining (immaterial):* a one-line collaborator confirmation that the MAT exclusion is still current guidance.

3. **PPFX effective rank — *resolved.*** The 100-universe flux band covariance has hard rank 99 but a participation-ratio effective rank of ≈ 1: a single coherent normalization mode carries 99.6 % of the variance (9 modes for 99.9 %). So the PPFX flux uncertainty is, to excellent approximation, a fixed-rank (rank-1, at most rank-9) object; treating the 100 universes as 100 independent directions over-counts the flux dimensionality. This is also why the leading flux mode correlates $\rho \approx 0.9$ with the rank-1 muon-energy-scale mode (item 1). A PCA / fixed-rank flux covariance is therefore both justified and recommended.

4. **Ours-only χ^2 — *method resolved; precedent open.*** The obstruction is statistical, not rank: $\mathbf{C}^{\mathrm{ours}} = \mathbf{C}^{\mathrm{syst}} + \mathbf{C}^{\mathrm{boot}} + \mathbf{C}^{\mathrm{ML}}$ is rank 201 and invertible; we report the ours-only goodness-of-fit on it via a truncated-spectral χ^2 (retain $\lambda > 10^{-10}\lambda_{\mathrm{max}}$), the same convention as the 3D comparison, and lead with per-bin pulls plus the regularization scan (§3.3). *Remaining:* published precedent from other MAT-based OmniFold/unbinned analyses for the ours-only χ^2 convention.

5. **3D systematic covariance — *method resolved; precedent open.*** The available-energy extension carries a full combined covariance (universe+bootstrap+ML) over all 1431 reported bins, rank 247 (§6.6); the goodness-of-fit is the rank-247 truncated-spectral χ^2 (§6.5), the recommended treatment for a rank-deficient unbinned covariance. *Remaining:* whether there is MINERvA precedent for *publishing* a 3D systematic covariance from low-recoil samples, and endorsement of this GoF.

6. **High-E_{avail} DIS-tail excess — *localized in W, generator- and tune-robust; modelling open.*** Enabling Valencia 2p2h resolves the low-recoil excess but leaves a separate $+2.2\sigma$ data excess in the $[1.50, 3.00)\,\mathrm{GeV}$ bin (§6.7), where 2p2h does not contribute and the (sub-percent) FSI dials of item 2 cannot reach. To test the deep-inelastic hypothesis directly we added the hadronic invariant mass W as a validated fifth OmniFold axis (its W-marginal recovers the frozen 4D cross section to $0.11\,\%$) and compared the unfolded (E_{avail}, W) cross section to the GENIE central value. The excess is *predominantly deep-inelastic*: the two highest-E_{avail} bands carry $57\,\%$ of the positive excess; of all positive excess, high-E_{avail} ($\geq 0.8\,\mathrm{GeV}$) holds $67\,\%$ and $83\,\%$ of *that* sits at $W \geq 1.8\,\mathrm{GeV}$, with the single largest cell the deep-DIS corner ($E_{\mathrm{avail}} > 3\,\mathrm{GeV}$, $W > 3\,\mathrm{GeV}$) at $22\,\%$. A secondary low-W ($< 1.1\,\mathrm{GeV}$, QE-like) excess and a small Δ-region ($W \simeq 1.4$–$1.8\,\mathrm{GeV}$) deficit are also present. So the missing strength behaves like a deep-inelastic-tail *modelling* deficit, exactly the hypothesis the W axis was added to test. We confirm this against a *generator band*: independent spline-normalised (E_{avail}, W) predictions from GENIE CV, GENIE + Valencia 2p2h/MEC, NuWro, and GiBUU all underpredict the high-$E_{\mathrm{avail}}\times$high-W corner by $54\,\%$ to $58\,\%$ (data/generator = 1.54, 1.58, 1.56). Crucially, enabling Valencia 2p2h does *not* close it — it slightly *worsens* the corner ($1.54 \to 1.58$), since 2p2h populates low W — and the independent NuWro and GiBUU generators miss it by the same or a larger margin (GiBUU is the most deficient overall, $\sigma = 2.22 \times 10^{-38}\,\mathrm{cm}^2/\mathrm{nucleon}$); at $W \in [2.2, 3.0)\,\mathrm{GeV}$ all four sit $23\,\%$ to $25\,\%$ below the data. The DIS-tail excess is therefore *generator- and tune-robust*: no current model reproduces it. Attaching a number, we marginalise the adopted combined 4D covariance (§6.6, unified-throw systematic) to the E_{avail} axis and evaluate the full-covariance χ^2 of the unfolded data against each generator's $d\sigma/dE_{\mathrm{avail}}$: all four are incompatible with the data at $> 21\sigma$ over the seven E_{avail} bins and $> 15\sigma$ in the DIS tail ($E_{\mathrm{avail}} \geq 0.8\,\mathrm{GeV}$) alone — GENIE CV 22.4σ, GENIE + MEC 24.9σ, NuWro 21.9σ, GiBUU 21.2σ. GiBUU spreads its deficit across the *entire* DIS tail (data/generator 1.59 at $E_{\mathrm{avail}} \in [0.8, 1.5)$, 1.36 at $[1.5, 3.0)$) rather than piling it in the deep-DIS catch bin like the GENIE variants, an independent and qualitatively distinct confirmation. Resolving the excess in W confirms and sharpens this picture: we build the full (E_{avail}, W) covariance (42 bins; `C_syst`+`C_stat`+`C_lateral`, median $14.8\,\%$ per bin) by a frozen-reweighter block sum on the 5D `_universes_full` sample, validated by reproducing the frozen 5D marginals to $0.1\,\%$. Over the full (E_{avail}, W) plane the four generators are incompatible with the data at GENIE CV 16.7σ ($\chi^2/\mathrm{ndf} = 412.7/42$), GENIE + MEC 16.1σ ($390.5/42$), NuWro 31.2σ ($1148.4/42$), and GiBUU $> 37\sigma$ ($1930.2/42$). The deficit is *localised in the high-W DIS corner*: restricting to the twelve bins with $E_{\mathrm{avail}} \geq 0.4\,\mathrm{GeV}$ and $W \geq 1.8\,\mathrm{GeV}$, all four miss the data at 9.0–18.2σ — GENIE CV 9.0σ ($116.9/12$), GENIE + MEC 9.2σ ($121.1/12$), NuWro 10.5σ ($149.6/12$), and GiBUU 18.2σ ($381.1/12$, again the most deficient). The excess is thus a genuinely DIS-region ($W \gtrsim 1.8\,\mathrm{GeV}$) feature, not a low-$W$ resonance artefact. *Remaining:* the cross-check against other MINERvA inclusive low-recoil samples [46].

A Statistical methods

The full uncertainty-quantification methodology — bootstrap, ML-seedscan, universe covariance construction (MAT-conformant `MnvVertErrorBand::CalcCovMx`), the $1/\Phi$ flux normalisation, the ill-conditioning/rank analysis, and the tension anatomy — is the companion document `docs/uq_statistical_meth` which this note treats as authoritative and does not duplicate.

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